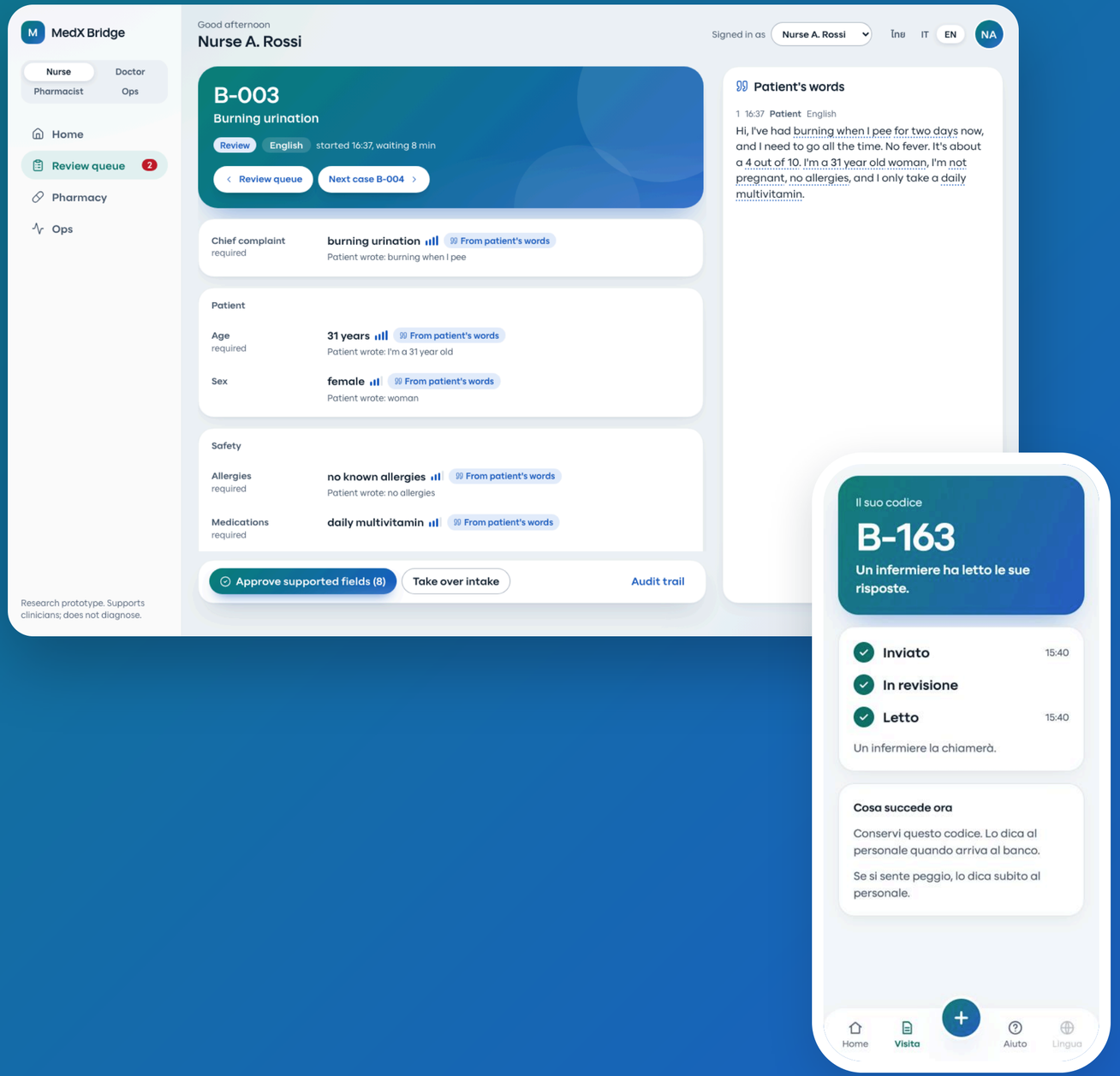
MedX Bridge

AMD Developer Hackathon ACT III · Track 2

# MedX Bridge

Multilingual intake for nurses and pharmacists. Every field points back to the patient's own words.




Research prototype. Supports clinicians; does not diagnose.
Synthetic data only. github.com/KingPhuripol/medx-bridge

# A language gap and a wrong medicine list are both safety events

**49.1% vs 29.5%**

**Physical harm in adverse events**, limited-English patients vs English speakers, six US hospitals. Communication errors caused 52.4% vs 35.9%.[1]

**13–59%**

**Patients with at least one unintended error** in their admission medication list across six studies; omission was the commonest type.[2]

**49.2% vs 27.0%**

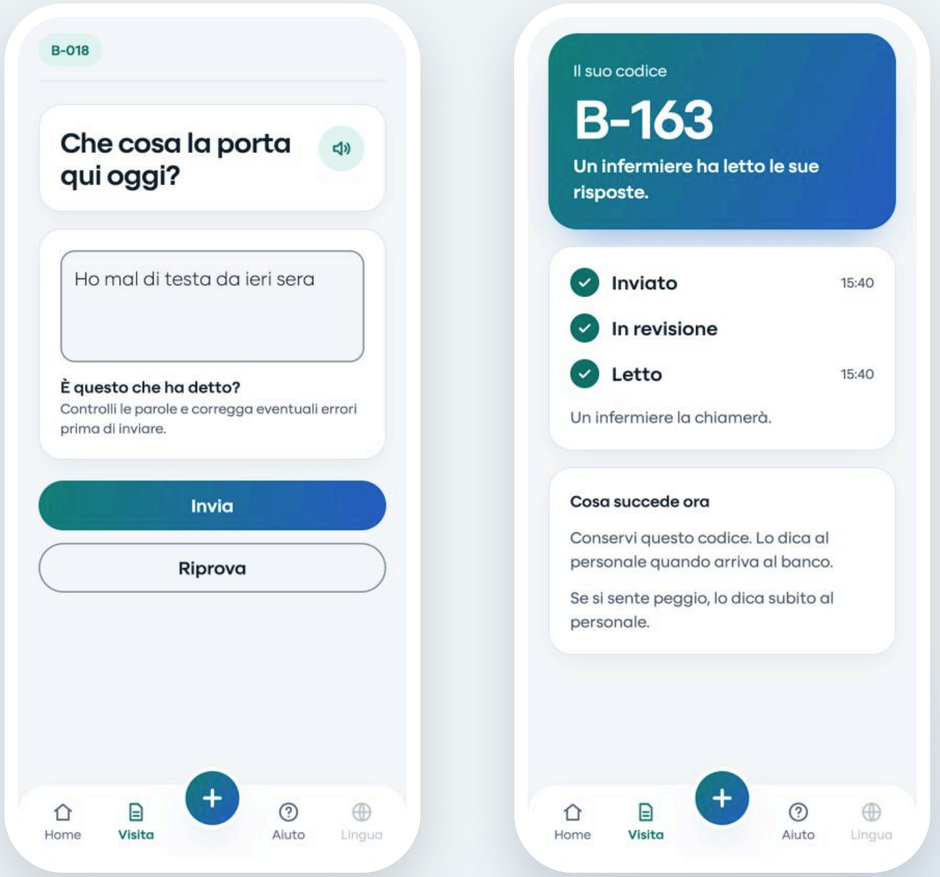
**Doctor's day on EHR and desk work** vs face time with the patient, four specialties.[3]

Thailand had 32.97 million foreign tourist arrivals in 2025.[4] At the desk, the patient and the staff often do not share a language.

1 Divi et al., Int J Qual Health Care 2007 (PMID 17277013). 2 Studies M1 to M6 in docs/research/PHARMACY-PROBLEM.md; settings differ, so a range is shown. 3 Sinsky et al., Ann Intern Med 2016 (PMID 27595430). 4 Ministry of Tourism and Sports via Bernama, 6 Jan 2026. Full list: docs/research/PROBLEM.md.

# Three people, one intake, before the consult

| | |
|---|---|
| **Patient** | Speaks or types in Thai, Italian or English. A voice agent reads the question aloud; the patient checks and corrects the transcript before it is sent. |
| **Nurse** | Reads an English summary (design goal: quick to scan). Clicks a field to see the patient's own sentence. Approves, edits or takes over. |
| **Pharmacist** | Reconciles the medicines the patient named, brand to generic, with alerts on duplicate paracetamol, allergy conflicts and interactions. |
| **Clinic manager** | Sees the queue by urgency, escalations, abstain rate and model latency. |




Left: the patient confirms what the voice agent heard. Right: after the nurse approves, the patient's phone ticks "Reviewed" in the patient's language.

Real screenshots from the built app (Expo patient app on web, Italian UI). "2 to 3 minutes per intake" is a design assumption, not a cited figure.

# Rules first, Gemma second, a human last

**1 Redact**
Names, phone, ID become tokens before any storage or model call.
code

→

**2 Red-flag rules**
18 rules, every turn. A hit stops the intake; no model can clear it.
code, no model

→

**3a Gemma on AMD**
Extract, translate, one quote per field.
model

**3b Gemma safety screen**
Add-only: raises, never clears.
model, in parallel

→

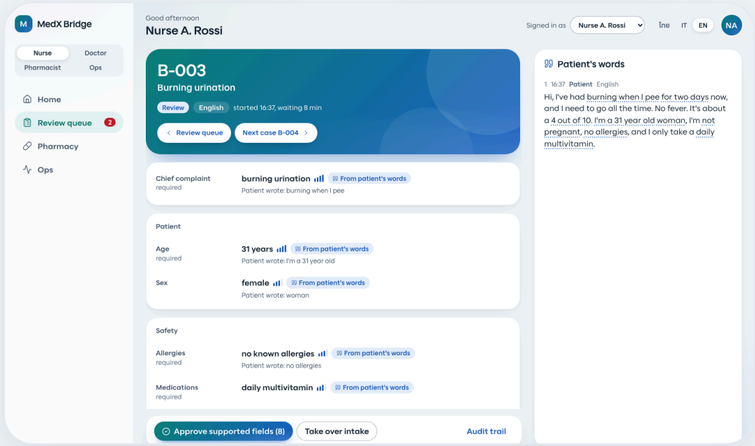
**4 Verify**
Code checks each quote exists. None: "needs check".
code

→

**5 Ask or hand over**
One follow-up question, or abstain.
model + code

→

**6 Clinician**
Approve, edit, take over. Audit log.
human




## From patient's words

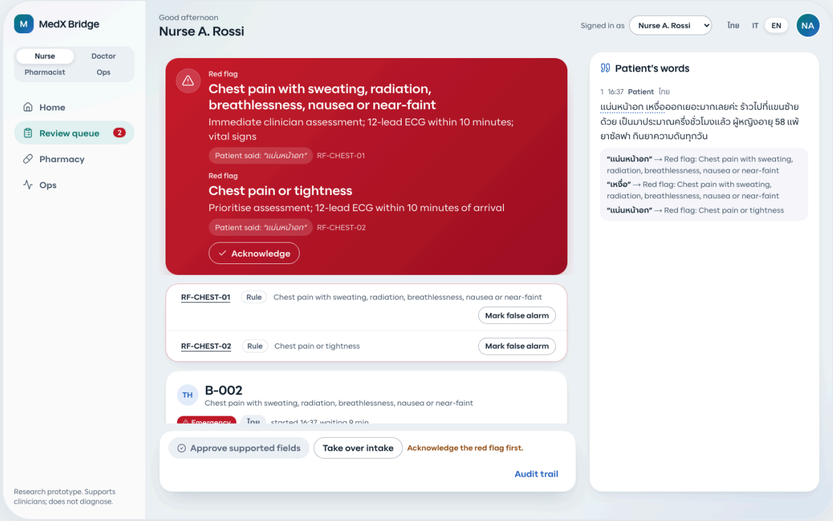
Each field in the nurse view carries the quote it came from, in the patient's own words, and a confidence bar. The highlighted sentence is the proof. Fields without a quote are shown as "needs check", never as fact.

Provider error, schema error or low confidence gives "needs human review", not a guess.

Design: docs/ai/DESIGN.md. Rules: docs/clinical/SPEC.md (18 rules, 175 reference tests). Screenshot: nurse review screen, synthetic case B-003.

# Safety by design, in five answers

| | |
|---|---|
| **Wrong or unsupported answers** | No verified quote means "needs check". Output is schema-checked; quotes must match the patient's words. Injection handling is not yet evaluated end to end. |
| **Sensitive information** | Redaction before storage or any model call. Synthetic data only. Keys stay server-side. |
| **Uncertain results** | Confidence is shown on every field. Low confidence or provider failure: abstain, "needs human review". |
| **High-risk decisions** | Rules run before any model. The Gemma screen is add-only, shown as "model-suggested, confirm". |
| **Human review** | Nothing is final until a clinician approves. Every action is in the append-only audit log. |




Thai patient, chest pain and sweating in the first message: the nurse screen is interrupted, the patient's sentence is highlighted, the nurse acknowledges or marks a false alarm with a reason. The patient phone shows a calm "go to the desk now" screen.

Mechanisms and code paths: pitch/SAFETY.md. Cross-vendor review: Claude read-only reviewed the Codex engine (docs/reports/w6-engine-review.md) and QA scripts hit every route (docs/reports/w7-qa-bugs.md); findings fixed with regression tests; remaining limitations are listed in the submission.

# Evidence we can show, and the gap that motivated the screen

| What | Count | Where |
|---|---|---|
| Synthetic cases (40 intake + 12 medication) | **52** | data/cases |
| Blind holdout cases, written without seeing the rules | **18** | data/holdout |
| Red-flag rules / reference tests | **18 / 175** | docs/clinical |
| Drugs in dictionary / reference tests | **59 / 194** | docs/clinical |
| Automated tests, all passing | **1,119** | apps/web, vitest |

Small synthetic blind holdout (18 cases), keyword rules alone

## 7 of 10

emergencies escalated; 3 of 8 non-emergencies raised a false alarm. Misses were lay wording that keyword lists cannot cover. That is why Gemma adds a safety screen.

**With the Gemma screen**

Holdout recall
[pending AMD run]

Asserted-field precision
[pending AMD run]

Unsupported rate, verifier on vs off
[pending AMD run]

Synthetic evaluation, research prototype, small n; not clinically validated. Holdout result: eval/results/holdout-rules-only.md, recorded before any change. Test count: npx vitest run in apps/web, 17 Oct 2026. Pre-declared metrics: docs/ai/METRICS.md.

# Gemma via AMD Token Factory, behind one gateway

**Clients**

Patient app (Expo), clinician, pharmacist and ops web (Next.js)

→

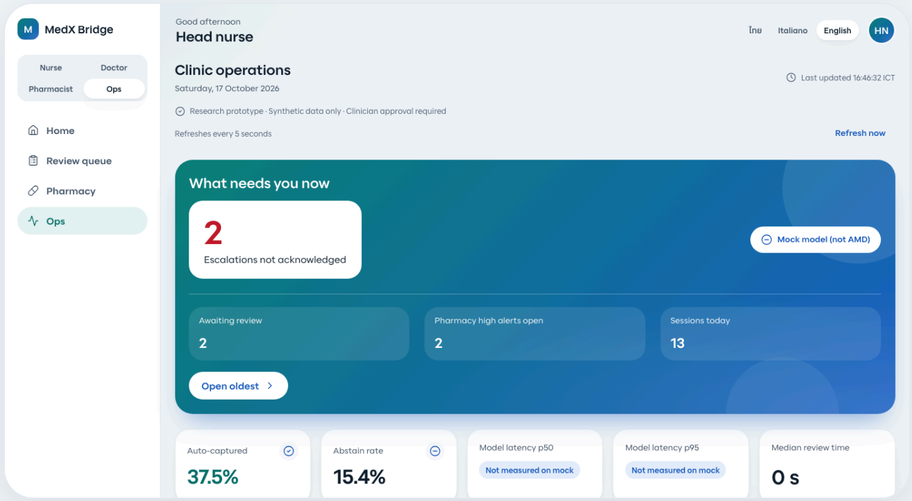
**API and pipeline**

Redact, rules, verify, audit. Provider SDKs stay inside one adapter.

→

**AMD Token Factory**

Designed to run Gemma on AMD. OpenAI-compatible; endpoint and model id come from environment variables.



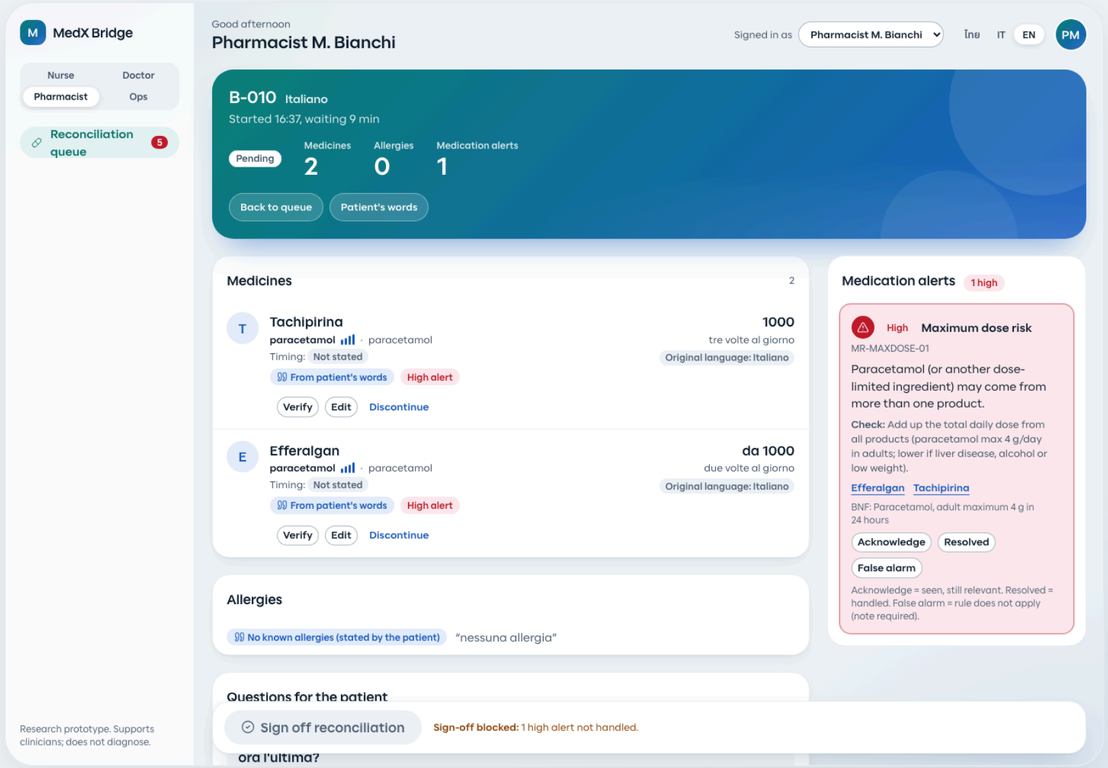

- Three jobs for Gemma (designed): extract and translate, add-only safety screen, next question.
- Current results use a mock model until the endpoint is connected; the ops screen says so ("Mock model, not AMD") instead of showing invented latency.
- Latency p50 / p95 per turn: **[pending AMD run]**
- Model variant chosen on a dev split: **[pending AMD run]**

Designed to run Gemma on AMD via Token Factory; the key was still pending, so current results use a mock model. Budget: turn p50 at most 6 s, p95 at most 12 s (docs/ai/DESIGN.md section 7), to be measured and reported either way.

# Next: prove it with clinicians before anyone relies on it

- Run the pre-declared evaluation on Gemma on AMD and publish every result, including misses.
- Native-speaker review of Thai and Italian text.
- Timed study with nurses and pharmacists, then ethics approval before any real data.
- Not a medical device. Supports clinicians; never diagnoses, prescribes or decides urgency.

Open source, MIT.
github.com/KingPhuripol/medx-bridge
Team: [names and roles to add]




Screenshot: pharmacist reconciliation, synthetic case B-010: Tachipirina and Efferalgan both contain paracetamol, flagged as a maximum-dose risk; sign-off is blocked until the alert is handled by the pharmacist.